Hello FPGA — Xilinx ISE setup on Ubuntu

Antoine C. · Follow
7 min read · Dec 18, 2020

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When it comes to FPGA design, you need to use a set of software to "compile" your description file and to program it through the JTAG port. However, while this software is often designed to run flawlessly on Windows, it is not always the case on Linux and more especially on YOUR Linux distribution.

In this article, I describe how to install, set up and flash a circuit (described in Verilog) on the Nexys 3 using ISE WebPack on Ubuntu (20.04).

Nexys 3

The Nexys 3 is a board built around a Spartan 6 FPGA chip from Xilinx. It can be programmed through a USB cable and feature several I/O interfaces such as USB, Ethernet or VGA. The FPGA itself contains 2278 slices, 576Kb of RAM, 32 DSP slice and an internal clock of 500MHz. We can also find a bunch of buttons, switches, LEDs and 7 segment display and some expansion port such as a VHDC connector and 4 Pmod ports. This information as well as the pinout can be found on the reference manual. However, since we use a Spartan 6 chip, the toolchain required is contained in ISE which is compatible with Windows 7 at most, SUSE Enterprise 11 and Red Hat Enterprise Workstation 5/6 as stated in this document. However, in 2020, we may want to run another distribution of Linux such as Ubuntu.

ISE WebPack

As stated earlier, ISE Webpack is not fully compatible with Ubuntu. To be fair, the main problem resides in the support of the cable driver. Therefore we can install the design suite with no problem.

ISE WebPack installation

You can find the software on the Xilinx website. You can select the 14.7 version then scroll to the ISE Design Suite and take care to download the Full Installer for Linux (You don't want to make a mistake when the download's size is about 6GB). You will need to create a Xilinx account.

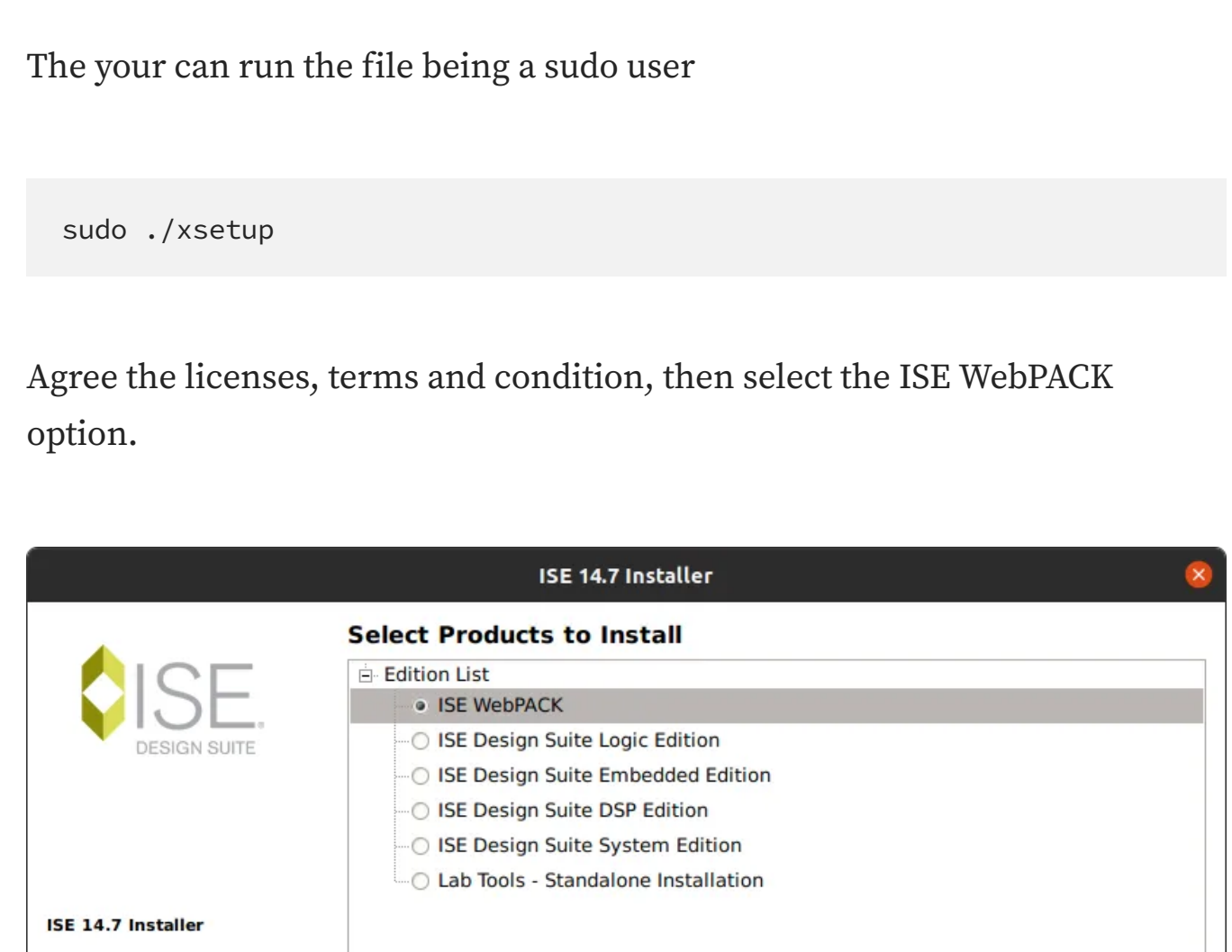
You can extract the TAR file wherever you want on your computer. You may need to add the execution right to the file xsetup

```
sudo chmod +x xsetup
```

The your can run the file being a sudo user

```
sudo ./xsetup
```

Agree the licenses, terms and condition, then select the ISE WebPACK option.



ISE 14.7 Installer
Select Products to Install
Edition List
● ISE WebPACK
○ ISE Design Suite Logic Edition
○ ISE Design Suite Embedded Edition
○ ISE Design Suite DSP Edition
○ ISE Design Suite System Edition
○ Lab Tools - Standalone Installation
Disk Space Required : 17403 MB
Description of ISE WebPACK
ISE 14.7 Installer
Welcome
Accept License Agreements
-> Select Products to Install
Select Installation Options
Select Destination Directory
Installation
< Back
Next >
Cancel

ISE WebPACK contains the most important tools you need for designing CPLDs and small to medium-sized FPGAs. Includes: ISE Design Tools (w/reduced device support), PlanAhead, Connectivity DSP IP, ChipScope Pro and The Embedded Development Kit will also be installed with WebPACK but are licensed separately (not included in a WebPACK license file).

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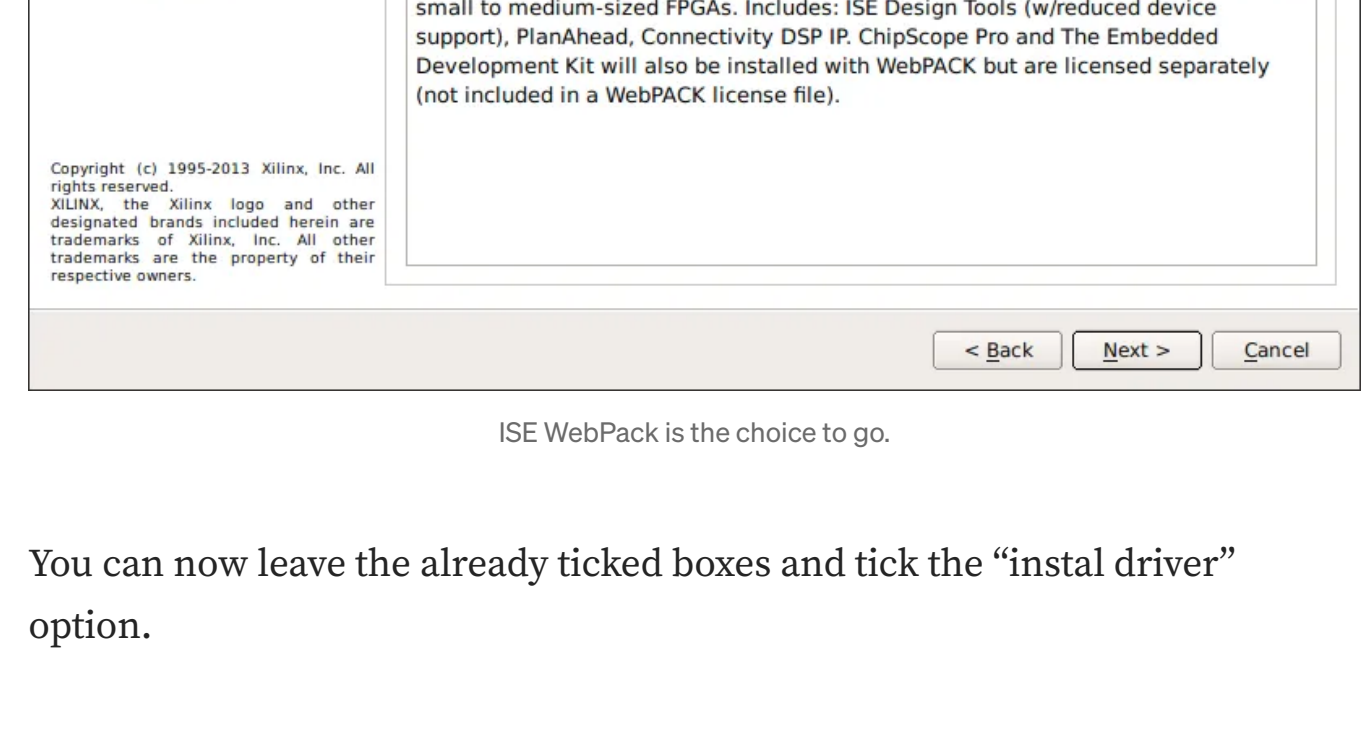
ISE WebPack is the choice to go.

You can now leave the already ticked boxes and tick the "instal driver" option.



ISE 14.7 Installer
Select Installation Options
ISE 14.7 Installer
Welcome
Accept License Agreements
Select Products to Install
-> Select Installation Options
Select Destination Directory
Installation
Select/Deselect All
Description of Install Cable Drivers
< Back
Next >
Cancel

Select the desired installation options below. Selection of these options may result in additional programs being run at the conclusion of the installation process.

☑ Use multiple CPU cores for faster installation

Enabling this option will speed up installation but may slow down other active applications.

☑ Acquire or Manage a License Key
☑ Ensure Linux System Generator Symlinks
☑ Enable WebTalk to send software, IP and device usage statistics to Xilinx (Always enal
☑ Install Cable Drivers

Cable drivers are required to ensure proper operation of the parallel and USB cables when configuring Xilinx devices. Please disconnect any Xilinx cables from your machine prior to driver installation.

Tick every boxes.

You can then continue with the location of the installation. The default one it in `/opt/Xilinx/14.7/ISE_DS` . It will bring you an error about the cable driver installation but the Design suite is installed. However, we can complete the installation by adding some scripts to get an icon as explained in this article.

Adding a shortcut

First, we're going to create a script creating a valid environment for ISE.

```
sudo nano /opt/Xilinx/14.7/ISE_DS/ise
```

Then, paste (`Ctrl + Maj + V`) the following content in the file :

```
#!/bin/bash
export LD_PRELOAD=/opt/Xilinx/usb-driver/libusb-driver.so
ISE_DS_DIR=/opt/Xilinx/14.7/ISE_DS
unset LD_PRELOAD
export gmake=/usr/bin/make

cd "$ISE_DS_DIR"
source "$ISE_DS_DIR"/settings64.sh

export LANG=''  # reset locale to English to fix decimal/comma
seperation
"$ISE_DS_DIR"/ISE/bin/lin64/ise
```

You now need to make this script executable :

```
sudo chmod a+x /opt/Xilinx/14.7/ISE_DS/ise
```

We just have to create the shortcut with a `.desktop` file:

```
sudo nano /usr/share/applications/ISE.desktop
```

Containing:

```
[Desktop Entry]
Version=1.0
Name=ISE
Exec=/opt/Xilinx/14.7/ISE_DS/ise
Terminal=false
Icon=/opt/Xilinx/14.7/ISE_DS/ISE/data/images/pn-ise.png
Type=Application
Categories=Development
```

Now, you can save the file and search for the application ISE in Ubuntu's research bar.

Cable Drivers

As you noticed you got some errors at the installation of the cable drivers. We are now going to solve this problem.

Cable driver install

First, we can install the support files as described by the above-cited article.

```
sudo apt-get install libusb-dev libftdi-dev build-essential libc6-dev
fxload

cd /opt/Xilinx
sudo git clone git://git.zerfleddert.de/usb-driver

cd usb-driver/
sudo make

./setup_pcusb /opt/Xilinx/14.7/ISE_DS/ISE/

echo PATH=\$PATH:/opt/Xilinx/13.2/ISE_DS/ISE/bin/lin64/ >> ~/.bashrc
echo export PATH >> ~/.bashrc
```

You might get some warning at the compilation, you can ignore them. However, these tasks alone are not sufficient.

Digilent's solution

You can install the Digilent's Adept 2 as suggested by attila on this forum answer. You will need to download the runtime and the utility on this page. You will not need the System and the SDK, only the Runtime and the Utilities are useful.

Once you downloaded them, install the `.deb` files:

```
sudo dpkg -i digilent.adept.runtime_2.21.2-amd64.deb
sudo dpkg -i digilent.adept.utilities_2.4.1-amd64.deb
```

> Note: You will need to modify the version by yours.

After the installation you can finally plug your board on the **USB PROG** and **Trun ON** the board with the **POWER switch** . Then run the command :

```
djtgcfg enum
```

You should see your device through the output :

```
Found 1 device(s)

Device: Nexys3
    Device Transport Type: 00010001 (USB)
    Product Name:          Nexys3
    User Name:             Nexys3
    Serial Number:         XXXXXXXXXXXX
```

Congratulation, your FPGA Board is now connected and operational!

A testing project

We are going to create a simple project using the switches 0, 1 and 2 to control the LEDs 0 to 7. The combination of the 3 switches gives a binary number turning on the corresponding LED. This part follows the tutorial of Jim Duckworth adapted to our version. Do not hesitate to follow this PDF to get more details.

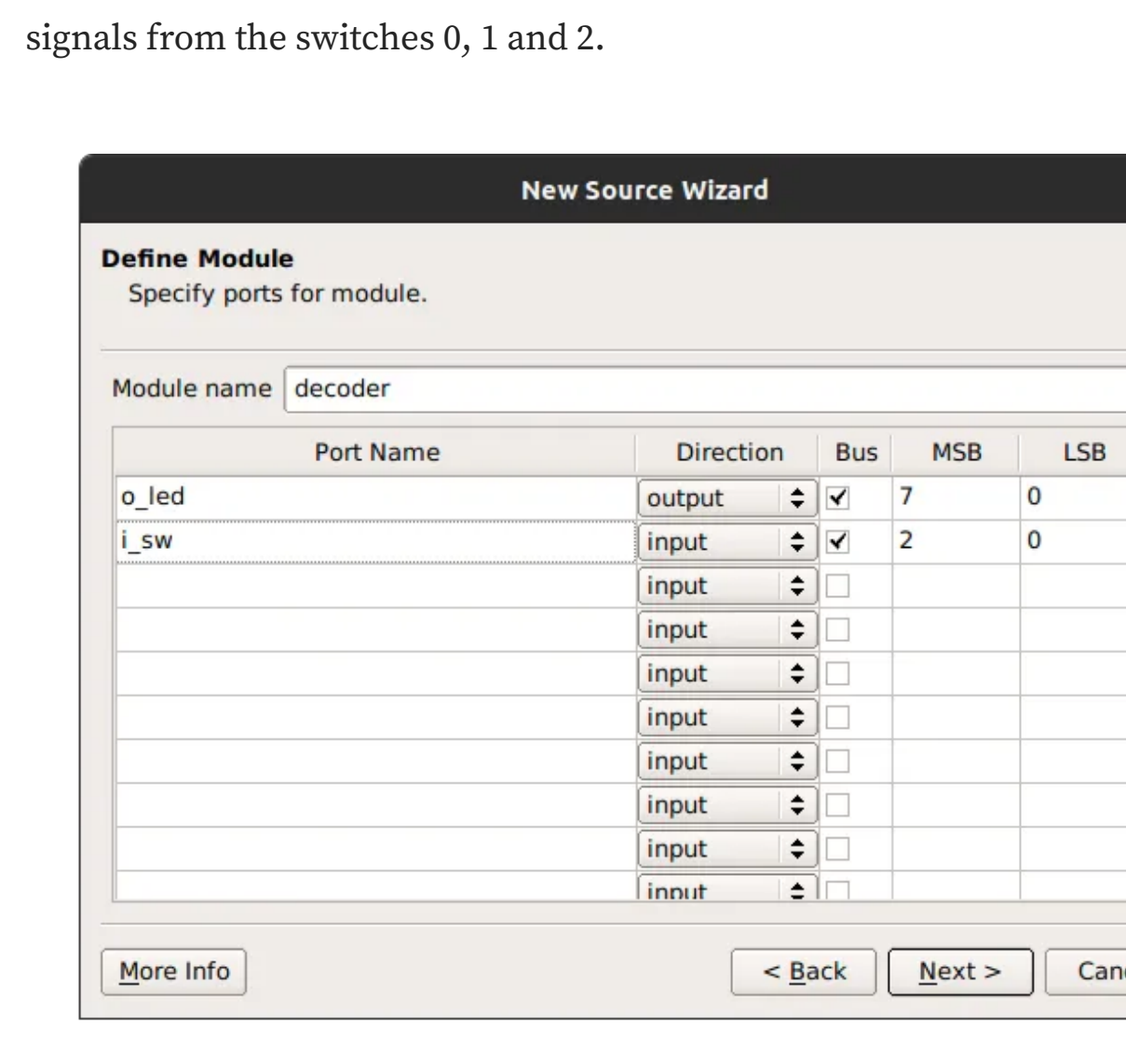
Creating the project

1. Start ISE and creat a new project with the name you want and the location you want. Next >
2. We can ignore the first value and focus on
   - Family: **Spartan6**
   - Device: **XC6SLX46**
   - Package: **CSG324**
   - Speed: -3 or -2
   - Synthesis Tool: **XST**
   - Similator: **ISim**
   - Prefered Language: **Verilog**
   Next >
3. A summary will appear at the screen, if everything is correct: Finish

Adding the code

On the left side bar, click on the New Source icon then select **Verilog Module**. We are gonna name it "decoder". Next >

We will **output** 8 signals on 8 LED from the LED0 to the LED7 and get 3 **input** signals from the switches 0, 1 and 2.



New Source Wizard
Define Module
Specify ports for module.
More Info
< Back
Next >
Cancel

Module name: decoder

Port Name	Direction	Bus	MSB	LSB
o_led	output	☑	7	0
i_sw	input	☑	2	0
	input	☐		
	input	☐		
	input	☐		
	input	☐		
	input	☐		
	input	☐		
	input	☐		
	input	☐		

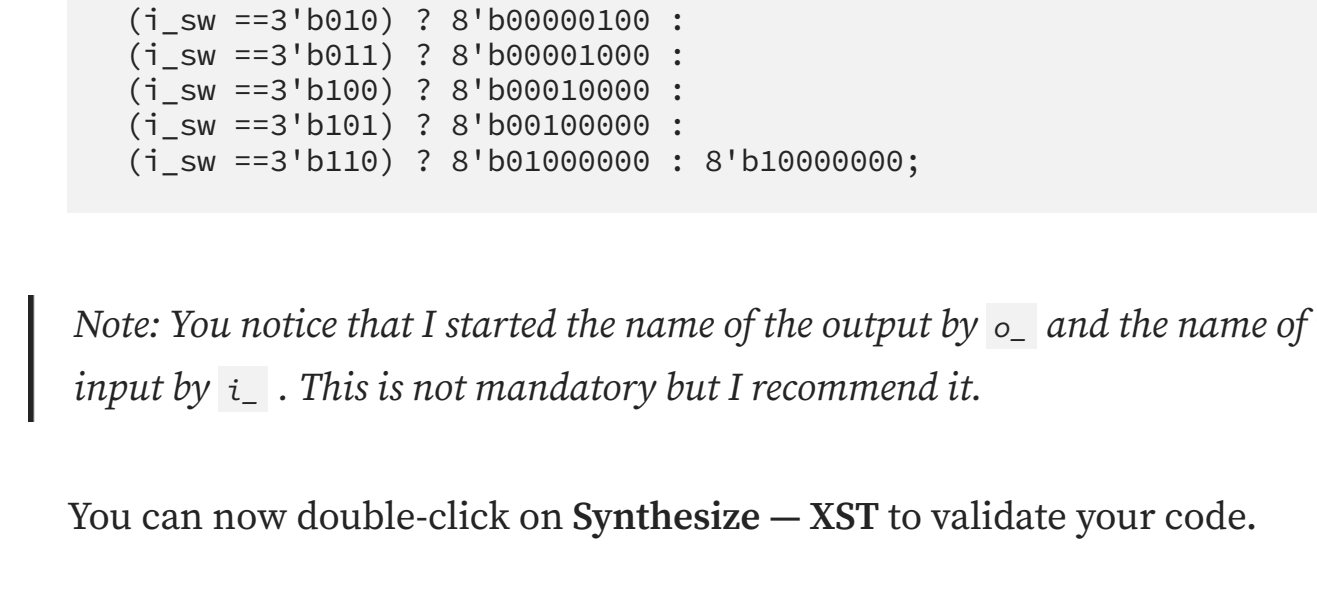
Defining the input and the outputs

We just need to add the following lines in the module definition (right before `endmodule`):

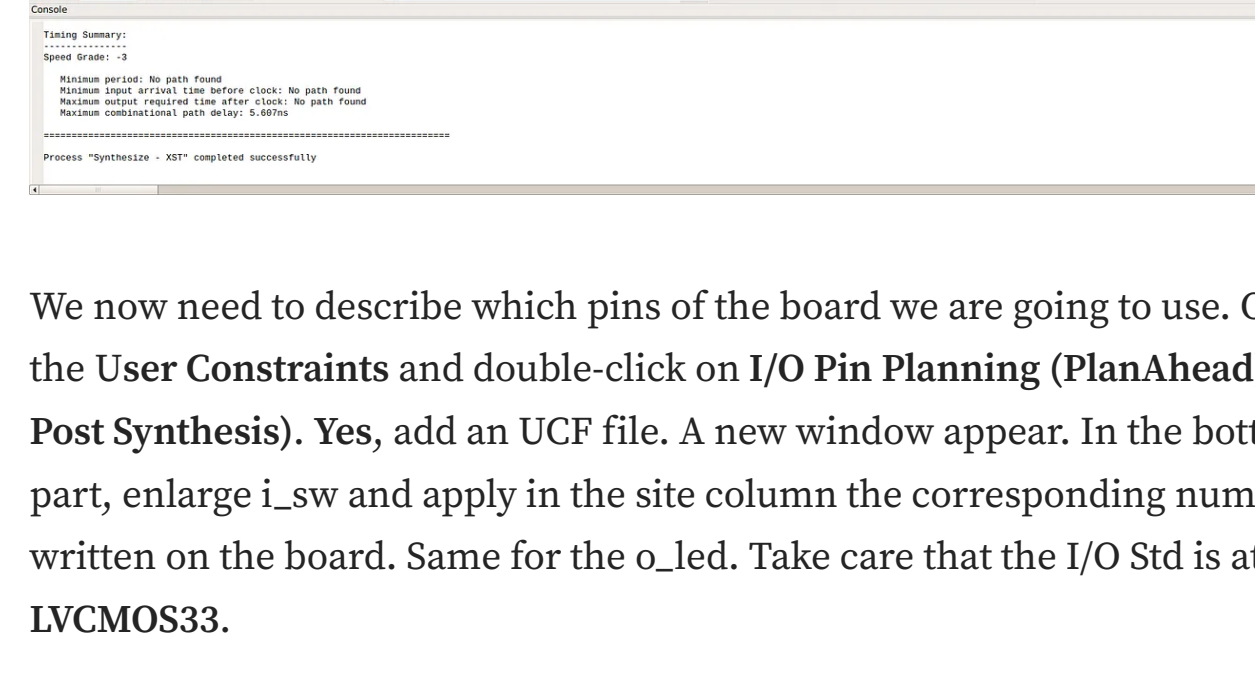
```
assign o_led = (i_sw ==3'b000) ? 8'b00000001 :
 (i_sw ==3'b001) ? 8'b00000010 :
 (i_sw ==3'b010) ? 8'b00000100 :
 (i_sw ==3'b011) ? 8'b00001000 :
 (i_sw ==3'b100) ? 8'b00010000 :
 (i_sw ==3'b101) ? 8'b00100000 :
 (i_sw ==3'b110) ? 8'b01000000 : 8'b10000000;
```

> *Note: You notice that I started the name of the output by `o_` and the name of the input by `i_` . This is not mandatory but I recommend it.*

You can now double-click on **Synthesize — XST** to validate your code.



We now need to describe which pins of the board we are going to use. Open the **User Constraints** and double-click on **I/O Pin Planning (PlanAhead) — Post Synthesis). Yes**, add an UCF file. A new window appear. In the bottom part, enlarge i_sw and apply in the site column the corresponding number written on the board. Same for the o_led. Take care that the I/O Std is at **LVCMOS33.**

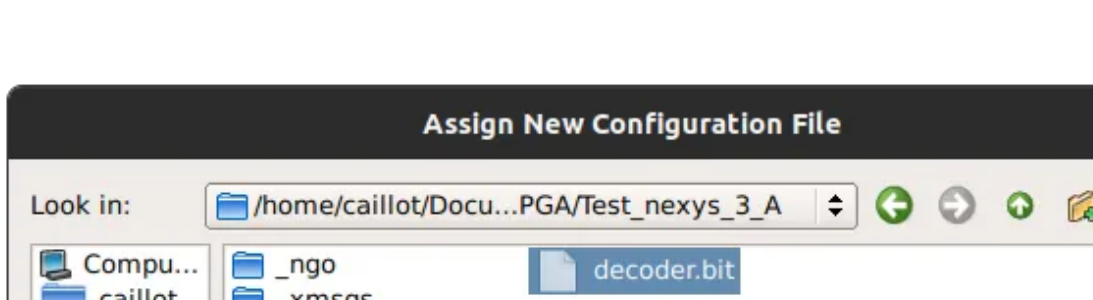


Final look

Save it (File > Save Constraints) and close the window. Back to ISE, Reload the files if it is asked.

Now, Double click to :

1. Implement design
2. Generate Programming File

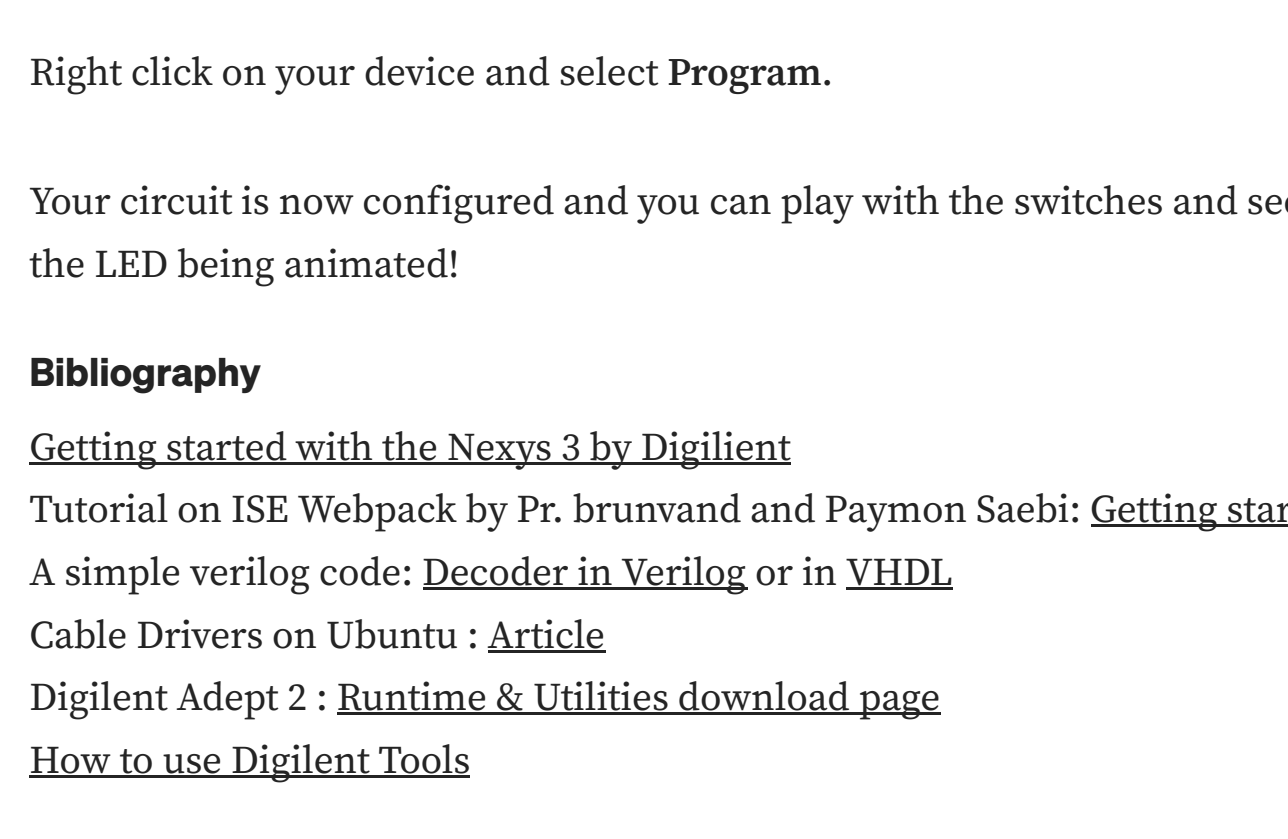


User Constraints
Create Timing Constraints
I/O Pin Planning (PlanAhead) - Pre-Synthesis
I/O Pin Planning (PlanAhead) - Post-Synthesis
Floorplan Area/IO/Logic (PlanAhead)
Synthesize - XST
Implement Design
Generate Programming File
Configure Target Device
Analyze Design Using ChipScope

Is it all green

If it is all green, all is good and we can continue. Otherwise, you need to correct the errors and the warnings displayed in the console.

Now it is time to flash or bit file. Double-click on Configure Target Device, OK the warning and welcome in a new window. From now on, we follow this tutorial by Pr. brunvand and Paymon Saebi. On the left side, double-click on **Bondary Scan** then right-click on the blue text in the middle: **Initialize Chain**. Your device should have appeared, you can press **Yes** and select the decoder.bit generated file. Press **no** on the Attach SPI or BPI PROM since we don't use them. Then press **OK** in the last window.



Assign New Configuration File
Look in:
/home/caillot/Docu...PGA/Test_nexys_3_A
Compu...
caillot
FPGA
_ngo
_xmsgs
ipcore_dir
iseconfig
nexys3_test_A_xdb
planAhead_run_1
planAhead_run_2
planAhead_run_3
xlnx_auto_0_xdb
xst
decoder.bit
File name:
decoder.bit
Files of type:
All Design Files (*.bit *.rbt *.nky *.isc *.bsd)
Open
Cancel
Bypass
Cancel All

Right click on your device and select **Program.**

Your circuit is now configured and you can play with the switches and see the LED being animated!

Bibliography

Getting started with the Nexys 3 by Digilent
Tutorial on ISE Webpack by Pr. brunvand and Paymon Saebi: Getting started
A simple verilog code: Decoder in Verilog or in VHDL
Cable Drivers on Ubuntu : Article
Digilent Adept 2 : Runtime & Utilities download page
How to use Digilent Tools

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Written by Antoine C.

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Former PhD Student working on cooperative perception for vehicles, now a postdoc fellow in Japan. Technology enthusiast, (astro)photography and RF aficionados.

Responses (1)

What are your thoughts?

Respond

Nkkilgupta
May 23, 2022 (edited)

On putting sudo ./xsetup

I am getting error

Downloads/Xilinx_ISE_DS_Lin_14.7_1015_1/bin/lin64/_xsetup: error while loading shared libraries: libncurses.so.5: cannot open shared object file: No such file or directory

What can be the reason?

1 reply Reply

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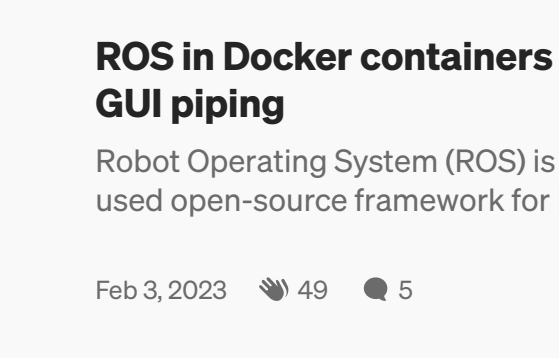

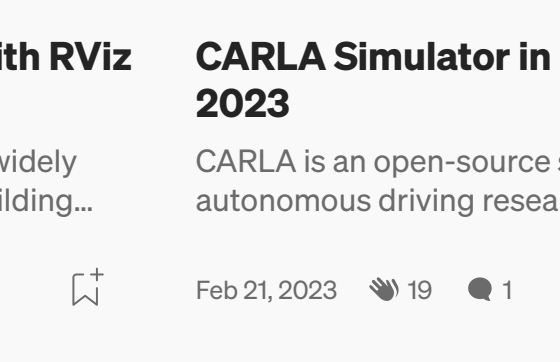

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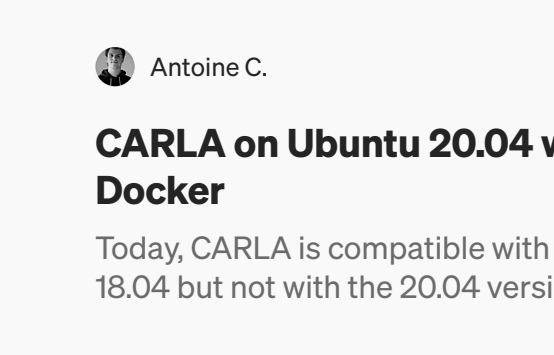

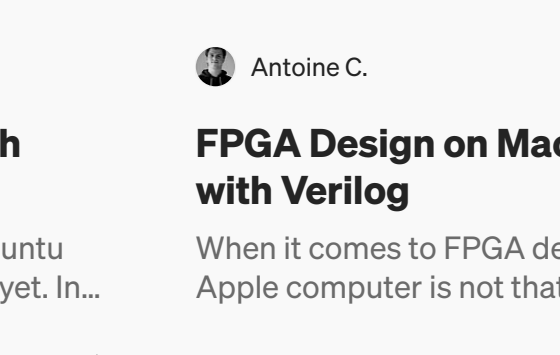

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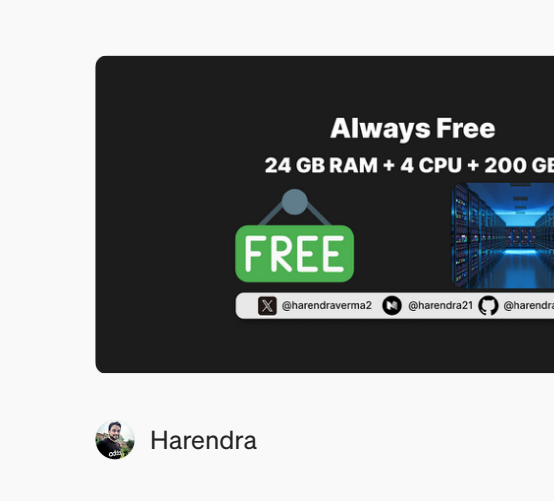

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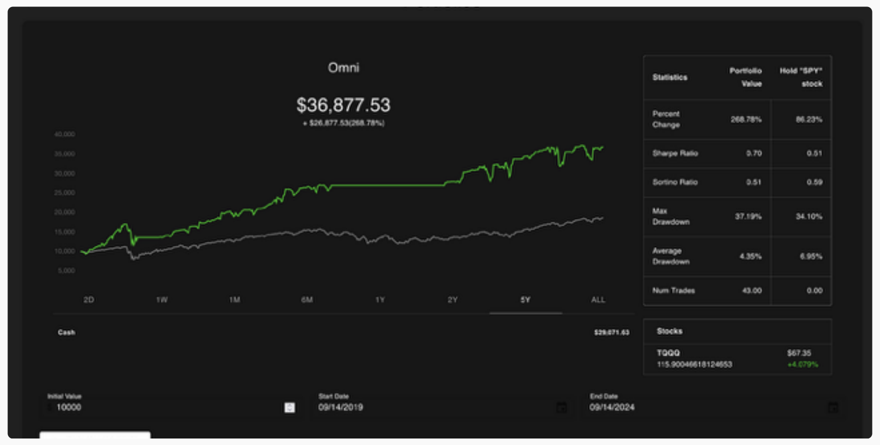

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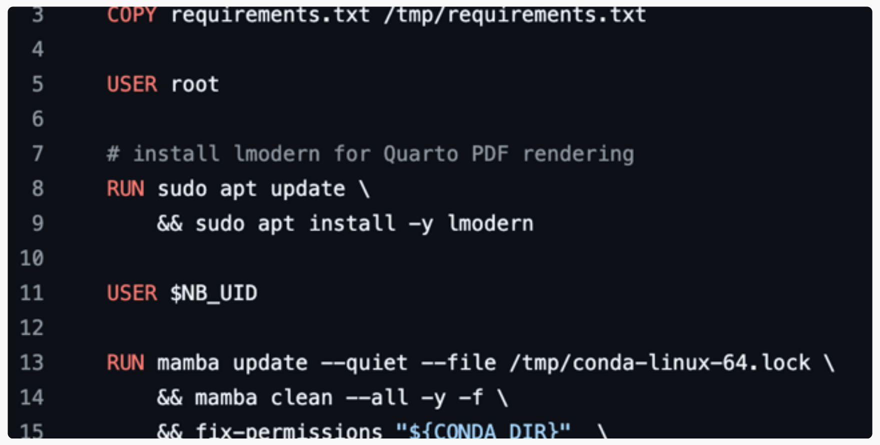

```
COPY requirements.txt /tmp/requirements.txt

USER root

# install lmodern for Quarto PDF rendering
RUN sudo apt update \
    && sudo apt install -y lmodern

USER $NB_UID

RUN mamba update --quiet --file /tmp/conda-linux-64.lock \
    && mamba clean --all -y -f \
    && fix-permissions "${CONDA_DIR}"  \
```

Daria Khon

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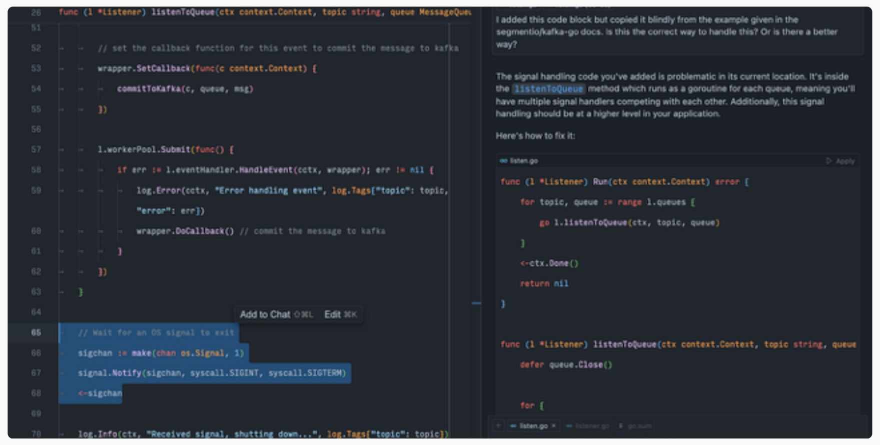

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